

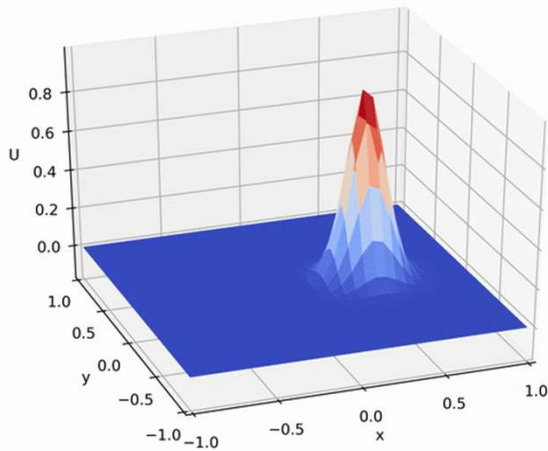


AIMS

African Institute for
Mathematical Sciences
SENEGAL

Numerical Optimization

Time = 0.000



Aurelle TCHAGNA, PhD, Eng

Senior Lecturer/ Researcher, Department of Computer Engineering

National Advanced School of Engineering, University of Yaounde 1

Researcher, Department of Medicine, McMaster University, Canada

aurelle.tchagna.univ-yaounde1.cm

tkaurelle@gmail.com

**Introduction of
Numerical Analysis**

**Numerical
Optimization in
Engineering**

**What is numerical
Method?**

AIM

**Applications of
Numerical
Optimization**

**Numerical
Optimization**

OUTLINE

Introduction to Numerical Analysis

□ Numerical Analysis

Numerical Analysis is the study of algorithms that use numerical approximation to well-defined mathematical problems.

Example 1

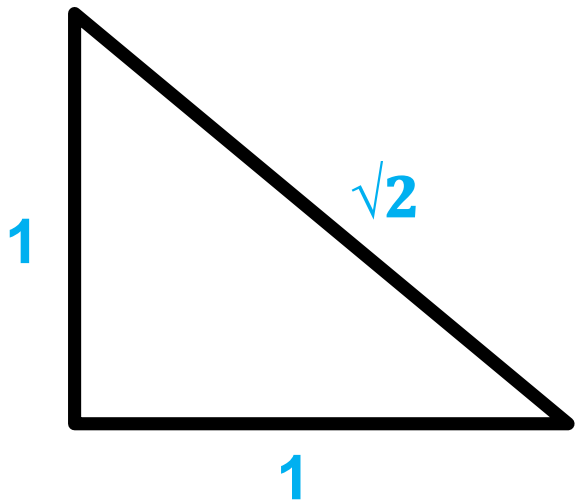
Find $x \in \mathcal{R}$ such that $x^2=2$.

This example tell us to find real number such that when it is multiplied by itself, the result is equal to 2.

The analytical solution for this problem is $x = \pm\sqrt{2}$

Where $\sqrt{2}$ is an irrational number.

□ Numerical Analysis



Pythagoras theorem is expressed as,

$$\text{Hypotenuse}^2 = \text{Base}^2 + \text{Height}^2$$

1.4142135624

For practical or engineering applications, an approximation in the form of a rational number that is « **close enough** » is used instead.

□ Numerical Analysis

❖ Babylonian Method

For Example, we could use the Babylonian method to find the value of x where $x * x$ is 2.

$$x \cdot x = 2 \quad \Rightarrow \quad x = \frac{2}{x} \quad \Rightarrow \quad x + x = x + \frac{2}{x}$$

$$2x = x + \frac{2}{x}$$



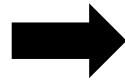
$$x = \frac{x + \frac{2}{x}}{2}$$

□ Numerical Analysis

❖ Babylonian Method

The approximation is found using an iterative method.

$$x = \frac{x + \frac{2}{x}}{2}$$



$$x_n = \frac{\left(x_{n-1} + \frac{2}{x_{n-1}}\right)}{2}$$

The first step in any numerical analysis is to assume something!

$$x_0 = 1$$

$$x_0^2 = 1 \neq 2$$

Introduction to Numerical Analysis

□ Numerical Analysis

❖ Babylonian Method

$$x_1 = \frac{\left(x_0 + \frac{2}{x_0}\right)}{2} = \frac{\left(1 + \frac{2}{1}\right)}{2} = 1.5$$

$$x_1^2 = 2.25 \neq 2$$

$$x_2 = \frac{\left(x_1 + \frac{2}{x_1}\right)}{2} = \frac{\left(1.5 + \frac{2}{1.5}\right)}{2} \approx 1.4167$$

$$x_2^2 = 2.007 \neq 2$$

$$x_3 = \frac{\left(x_2 + \frac{2}{x_2}\right)}{2} = \frac{\left(1.4167 + \frac{2}{1.4167}\right)}{2} \approx 1.41422\dots$$

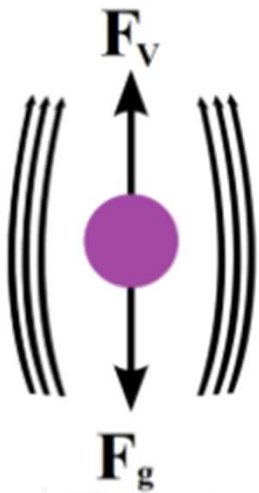
$$x_3^2 = 2.00002 \neq 2$$

NB : Stop after reaching a particular precision. We need to find a measure of the error in the approximate solution at each iteration

□ Numerical Analysis

Example 2

Find the velocity of a falling ball



Where

$F_v = -c\mathbf{v}$ (upward force of the air resistance)

$F_g = m\mathbf{g}$ (downward pull of gravity)

Newton's Laws ($\mathbf{F} = m\mathbf{a}$)

$$\frac{dv}{dt} = \frac{F_v + F_g}{m}$$



$$\frac{dv}{dt} = g - \frac{c}{m}v$$

Differential Equation
that describes the
phenomenon of
interest

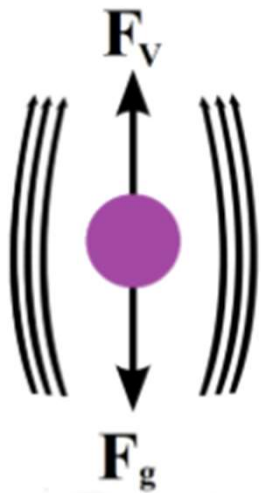
Introduction to Numerical Analysis

$$\frac{dv}{dt} = g - \frac{c}{m} v$$

□ Numerical Analysis

Example 2

Find the velocity of a falling ball



$$\frac{dv}{dt} = g - \frac{c}{m} v$$

We can solve this analytically using calculus (not arithmetic operations).

For example, if the ball was initially at rest ($v=0$ at $t=0$)

$$v(t) = \frac{gm}{c} (1 - e^{-(c/m)t})$$

Where

v = dependent variable

m, c = Parameters

t = Independent variable

g = Forcing Function

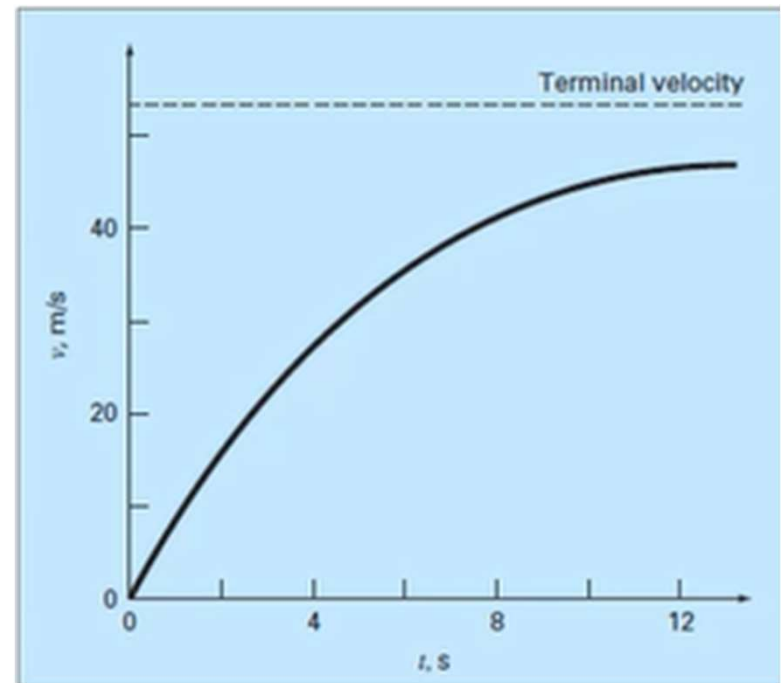
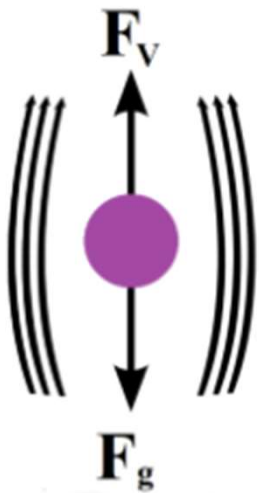
□ Numerical Analysis

Example 2

Find the velocity of a falling ball

$$v(t) = \frac{gm}{c} (1 - e^{-(c/m)t})$$

Plugging in appropriate values, we can find an analytical solution to this problem.



□ Numerical Analysis

Example 2

$$v(t) = \frac{gm}{c} (1 - e^{-(c/m)t})$$

We can solve this numerically by reformulating the mathematical problem so that it can be solved by arithmetic operations.

For example, the time rate of change of velocity can be approximated by:

$$\frac{dv}{dt} \approx \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

$$\text{Recall: } \frac{dv}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

Where

Δv = It is the difference in velocity over a finite interval

Δt = It is the difference in time over a finite interval

$v(t_i)$ = The velocity at a time t_i

$v(t_{i+1})$ = The velocity at a later time t_{i+1}

□ Numerical Analysis

Example 2

$$v(t) = \frac{gm}{c} (1 - e^{-(c/m)t}) \quad \Rightarrow \quad \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} = g - \frac{c}{m} v(t_i)$$



$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m} v(t_i) \right] (t_{i+1} - t_i)$$

□ Numerical Analysis

Example 2

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m} v(t_i) \right] (t_{i+1} - t_i)$$

This gives us a means to (algebraically) determine the velocity at time t_{i+1} using the previous values of velocity and time.

This new value of velocity at time t_{i+1} can be used to find the velocity at t_{i+2} and so on.

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m} v(t_i) \right] (t_{i+1} - t_i)$$

New Value = Old value + [Slope] x (Step Size)

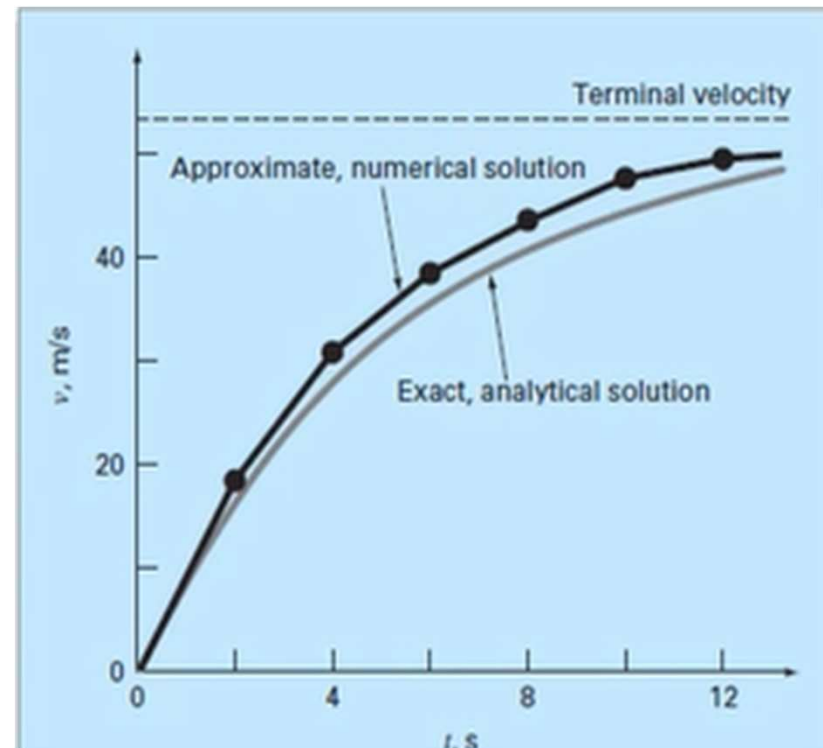
Introduction to Numerical Analysis

□ Numerical Analysis

Example 2

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m} v(t_i) \right] (t_{i+1} - t_i)$$

The approximate solution can be improved by, for example, decreasing the step size.



□ Numerical Analysis

Numerical Analysis is the study of algorithms that use numerical approximation to well-defined mathematical problems.

In numerical analysis, a **numerical method** is a mathematical tool designed to solve numerical problems.

What is Numerical Methods?

What are numerical methods?

In numerical analysis, a **numerical method** is a mathematical tool designed to solve numerical problems.

They are **algorithms** that compute **approximations** to functions, their derivatives, their integrations, and solutions to various equations.

Such algorithms could be implemented (programmed) on a computer.

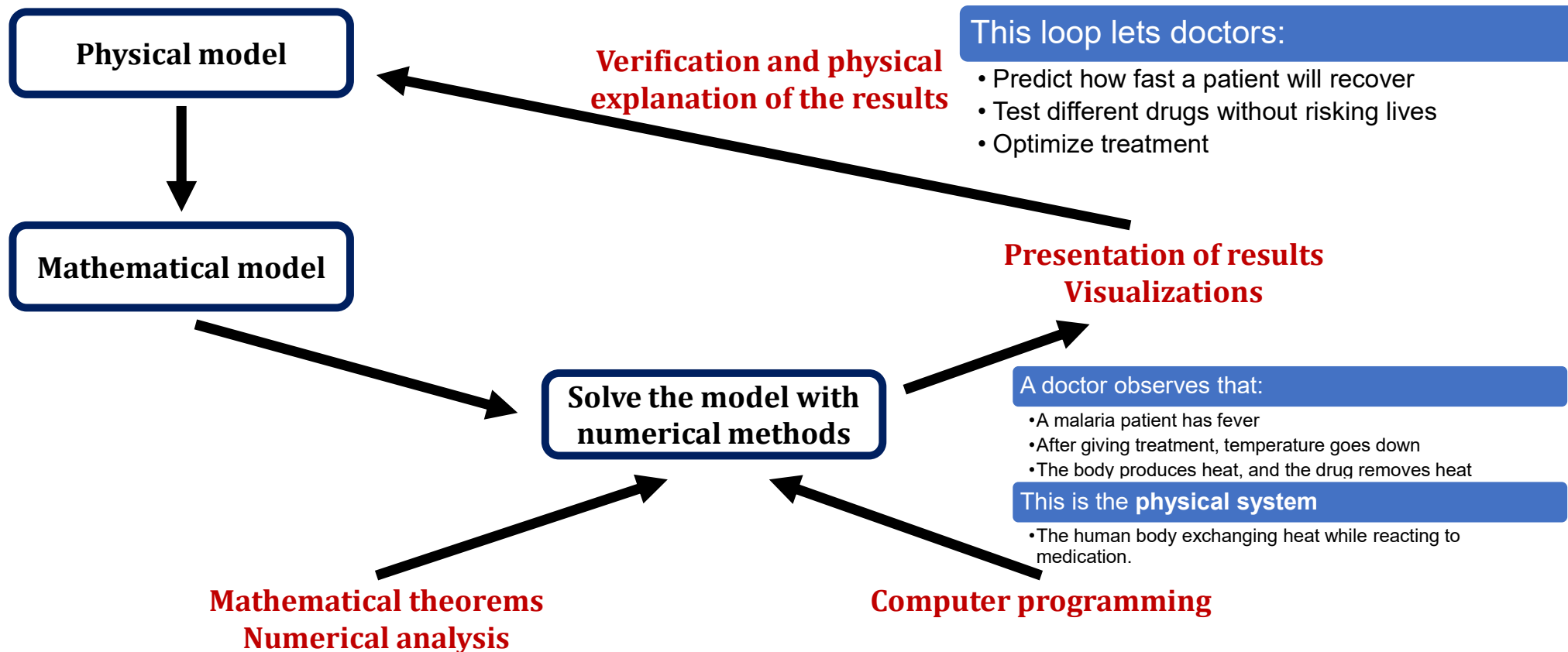
What is Numerical Methods?

❑ Need of numerical methods

- 1. Mathematical models are the central piece of science and engineering.**
- 2. Some models are closed from the solution, therefore they can be solve analytically. Many models can not be solve analytically or the analytical solution are too costly to be practical.**
- 3. All models can be solved computationally and the results may not be the exact answer but it can be useful.**

What is Numerical Methods?

Below is an overview on how various aspects are related



What is Numerical Methods?

□ Numerical methods

Steps involves to solve:

- ❖ Formulation of mathematical model;**
- ❖ Construction of an appropriate numerical method;**
- ❖ Implementation of the numerical method;**
- ❖ Obtain the solution by solving;**
- ❖ Validate the solution.**

What is Numerical Methods?

Keep in mind that numerical methods are not about numbers. It is about mathematical ideas and insight.

We will study some basic classical types of problems:

- ❖ Development of algorithms;
- ❖ Implementation;
- ❖ A little bit of analysis, including error-estimates, convergence, stability, etc.

We will use **Python** throughout the course for programming purpose.

Numerical Methods in Engineering

❑ Why are numerical methods used in Engineering

- 1. Engineers use mathematical modelling which includes various equations and data to describe and predict the behavior of systems;**
- 2. Computers are widely used which give accurate results and are cheap and affordable to all;**
- 3. Many software packages are available that can be used to solve problems.**

Numerical Methods in Engineering

□ How is it applied

- 1. Develop a mathematical problem with your skills and the requirements;**
- 2. Come out with a numerical algorithm;**
- 3. Implement the algorithm;**
- 4. Run, debug, test the code;**
- 5. Visualize and interpret the result;**
- 6. Validate the result.**

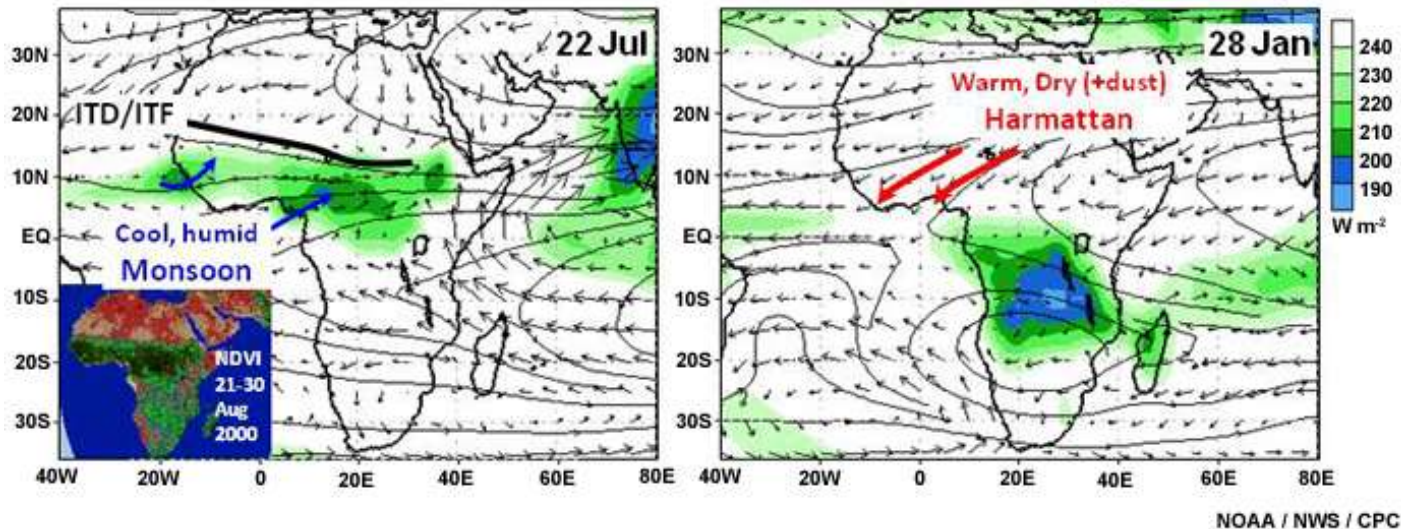
❑ Scientific Computing

- 1. Design and analysis for numerically solving mathematical problems in science and engineering;**
- 2. Effect of approximations and performs error analysis;**
- 3. Modern simulations of scientific applications;**

Applications of Numerical Analysis

☐ Making Weather Predictions

Meteorologists



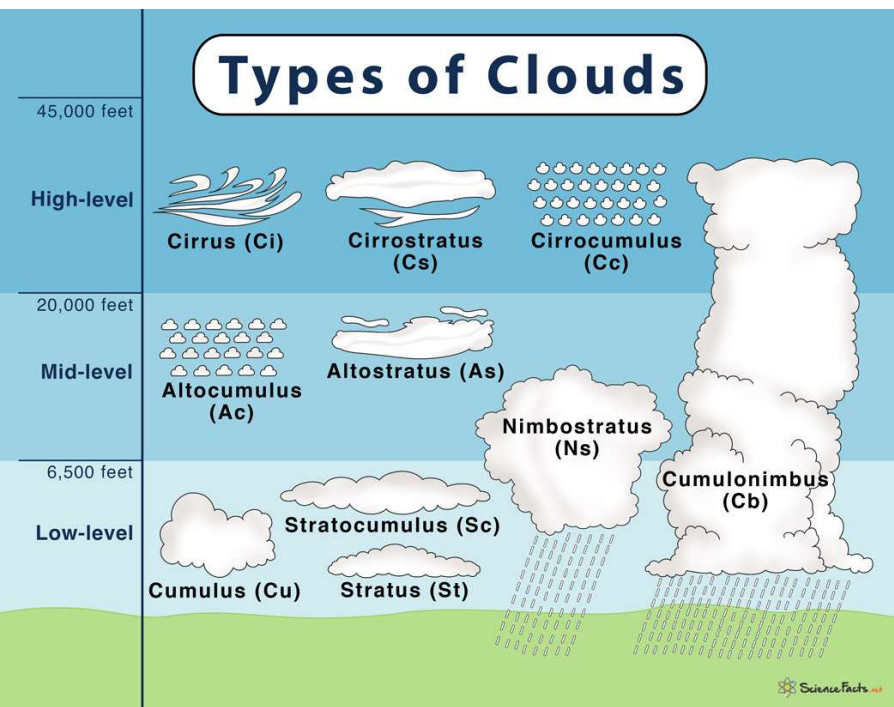
Meteorologist uses satellites, radars, and weather balloons to forecast the weather.

Applications of Numerical Analysis

❑ Making Weather Predictions

Study the weather conditions
using information obtained from:

- Clouds in the sky;
- Finding previous weather patterns;
- Current weather conditions;
- Tracking the motion of air.



Applications of Numerical Analysis

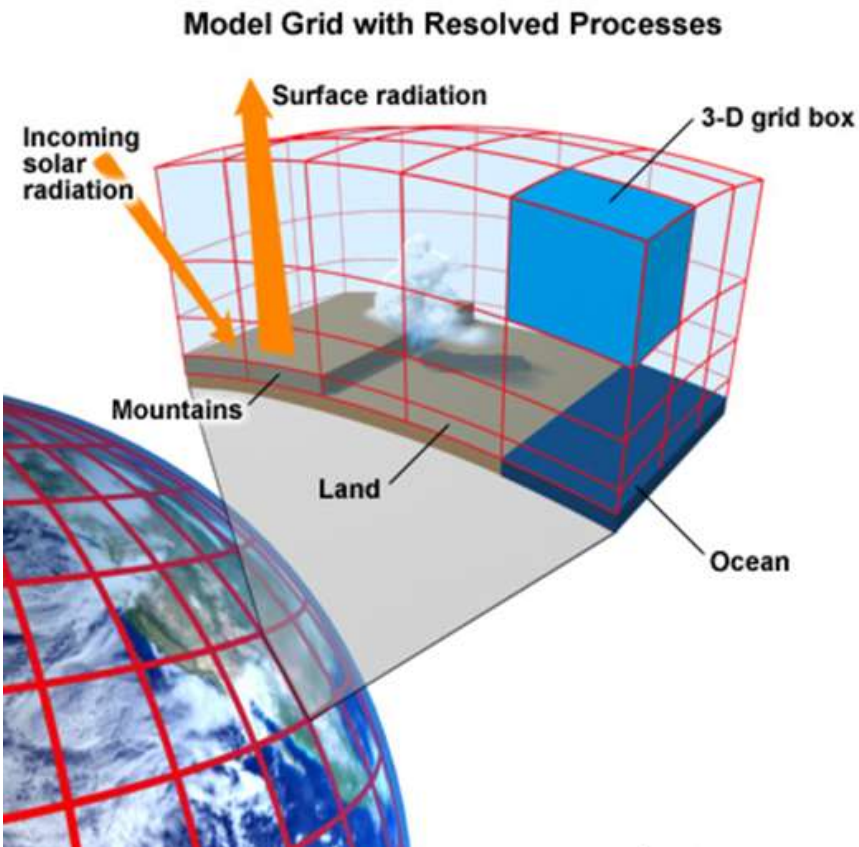
❑ Making Weather Predictions



Data is plugged into powerful super computers to create forecast models of the atmosphere.

Applications of Numerical Analysis

❑ Making Weather Predictions

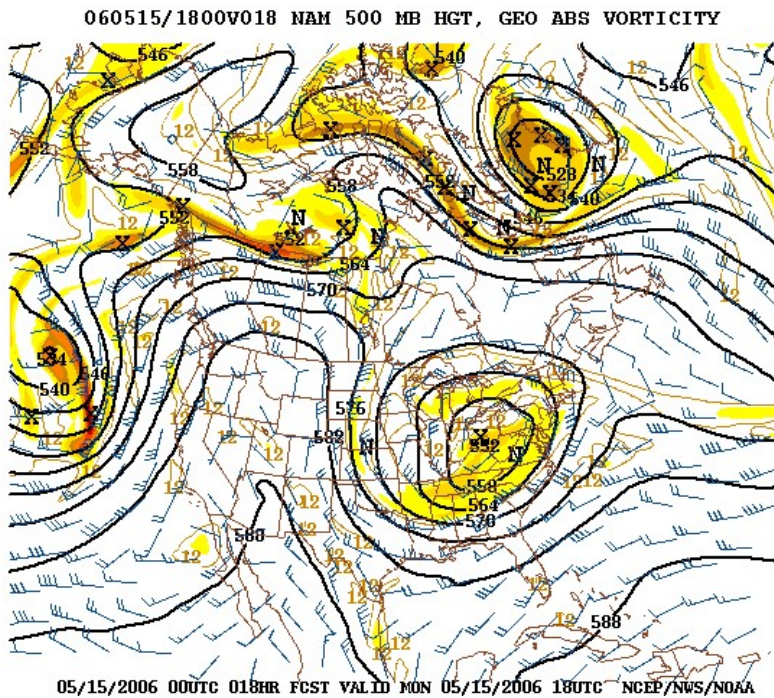


- ❖ Weather models use a coordinate system which divides the planet into a 3D grid.
- ❖ In each grid, heat transfer, solar radiation, phase change of water is calculated.

Applications of Numerical Analysis

❑ Making Weather Predictions

Weather models are simulations of the future state of the atmosphere through out time.

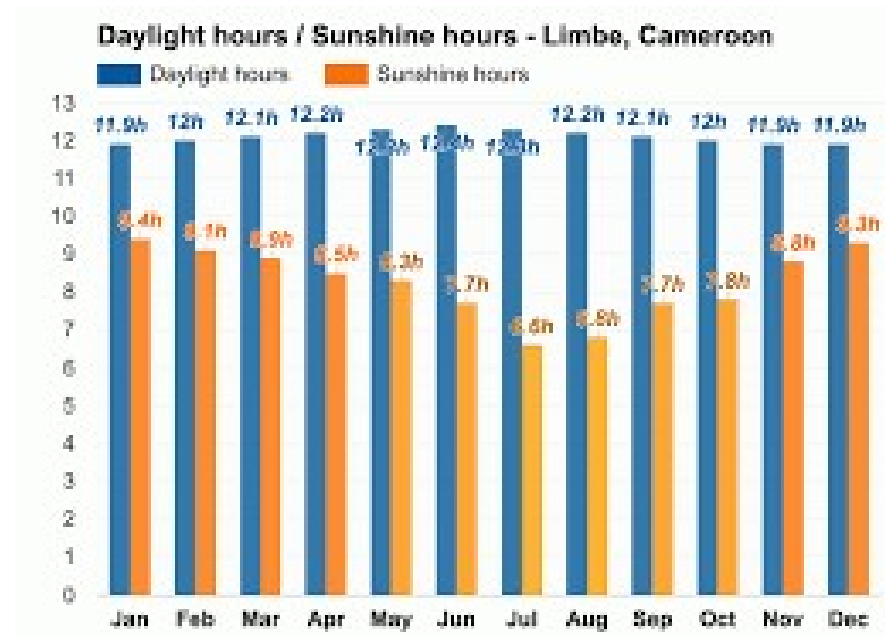


- ❖ An atmospheric model is a set of equations;
- ❖ These equations are non linear partial differential equations which are impossible to solve by analytical method.

Applications of Numerical Analysis

❑ Making Weather Predictions

- ❖ Therefore Numerical methods obtain approximate solutions;
- ❖ Different models use different methods but all the global models use **Finite difference methods**.



Applications of Numerical Analysis

❑ Making Weather Predictions



Natural Disasters



Agriculture



Transport



Air Traffic



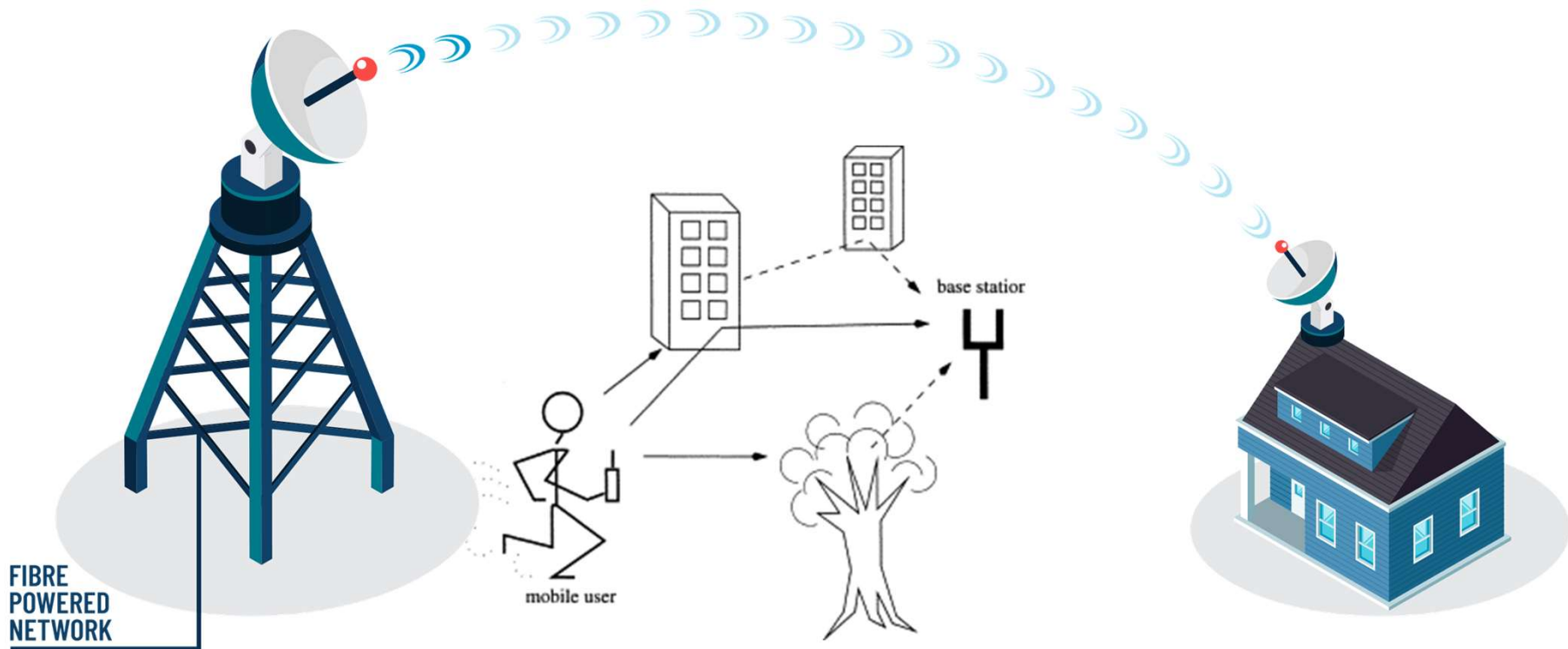
Private Sector



military

Applications of Numerical Analysis

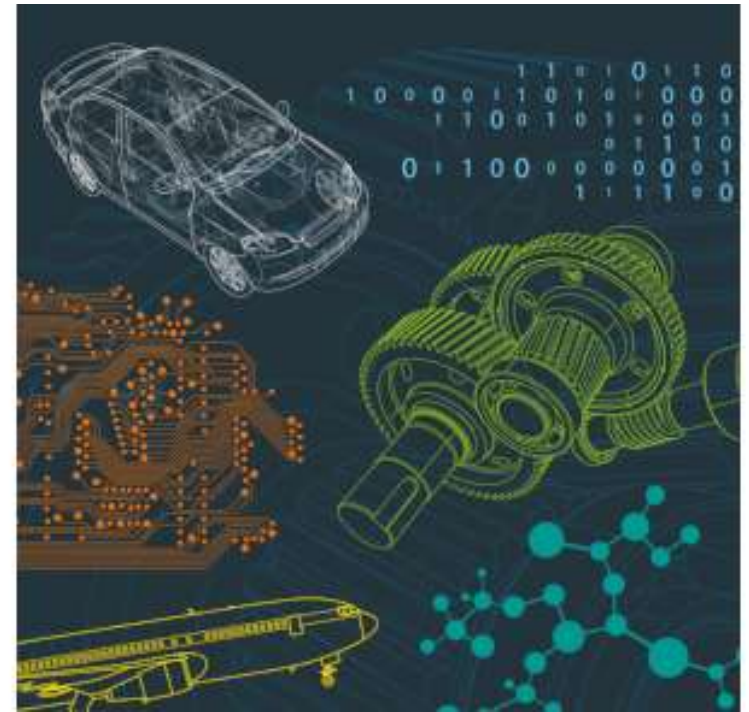
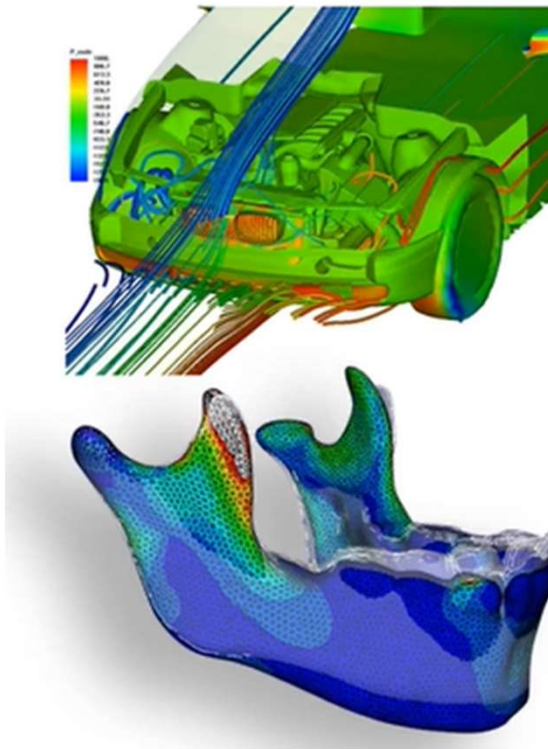
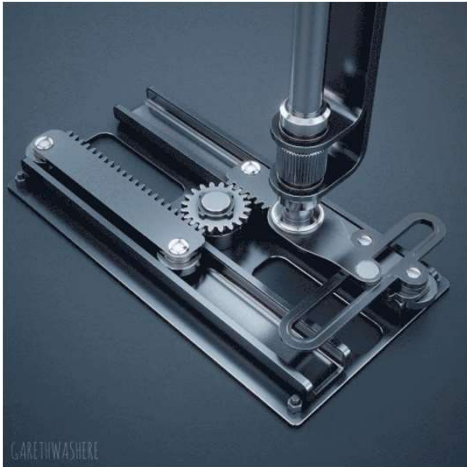
□ Telecommunications



- ❖ Maximizing the Sensibility of the signal for the Receiver;
- ❖ Minimizing the fading of the signal;
- ❖ Communication Network by Optimizing the Paths of the Signal.

Applications of Numerical Analysis

□ Mechanical Engineering



Applications of Numerical Analysis

❑ Finance



Financial risk

Financial structure

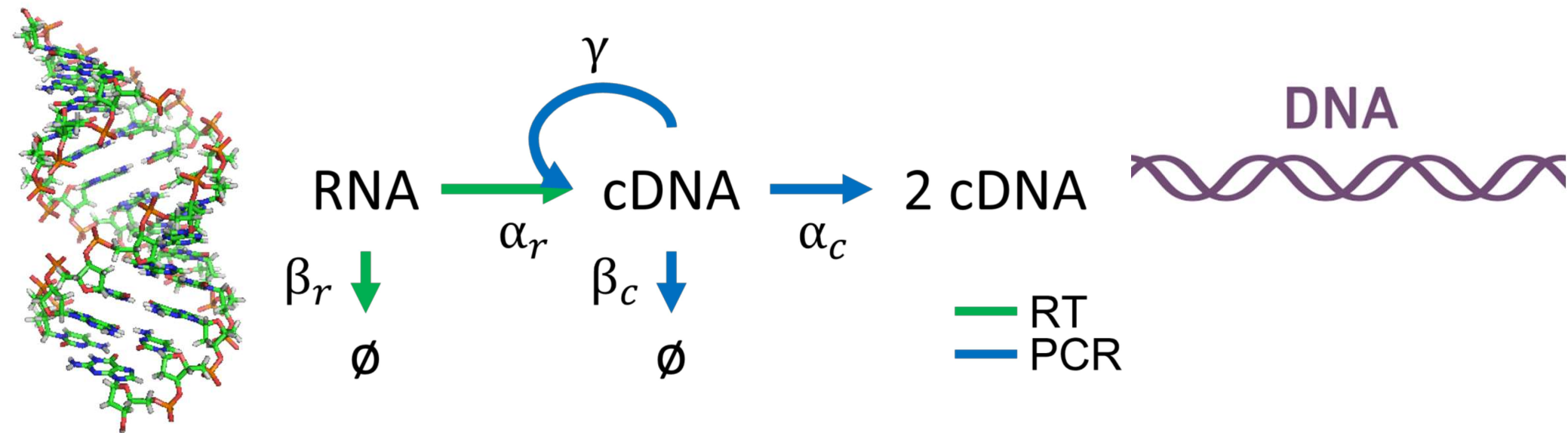
Market modeling

Capital budgeting

Track income and expenses

Applications of Numerical Analysis

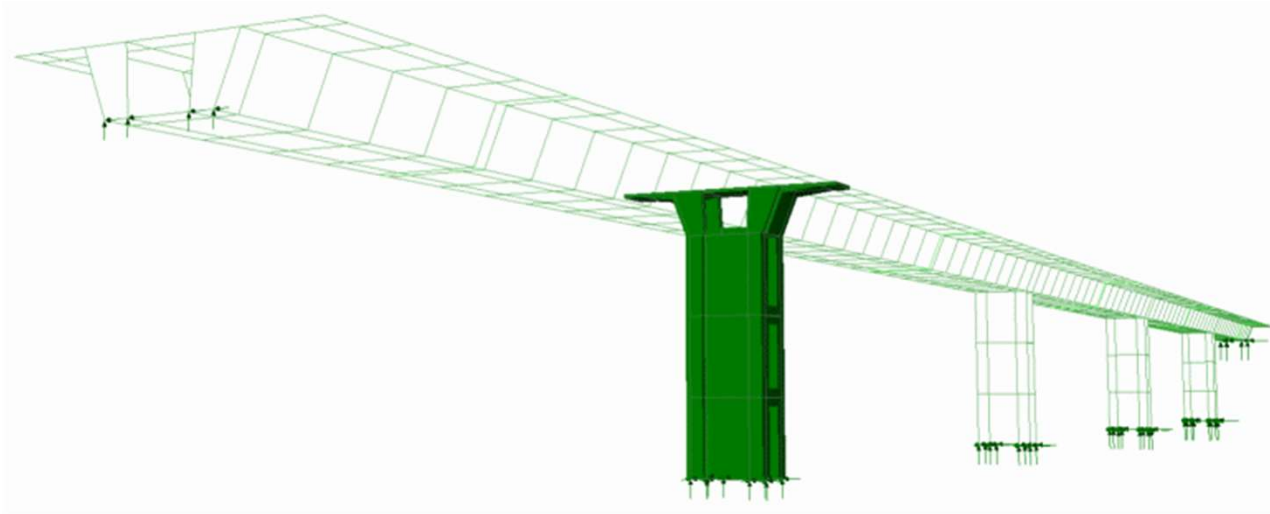
❑ Bio-Informatics



- ❖ Characterizing Genetic Sequences;
- ❖ Computational sequence analysis;
- ❖ DNA, RNA, and protein-sequence Data Analysis.

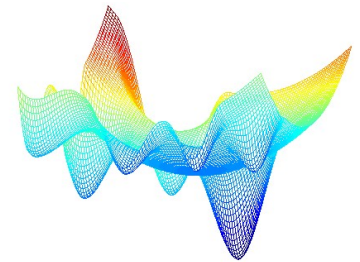
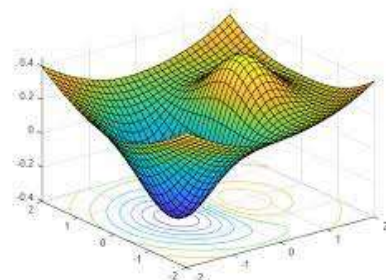
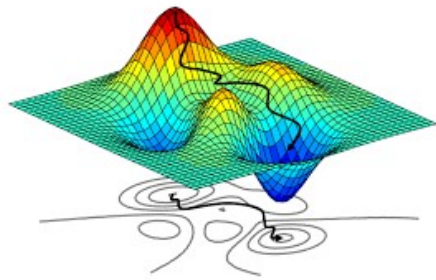
Applications of Numerical Analysis

□ Civil Engineering



Numerical Optimization

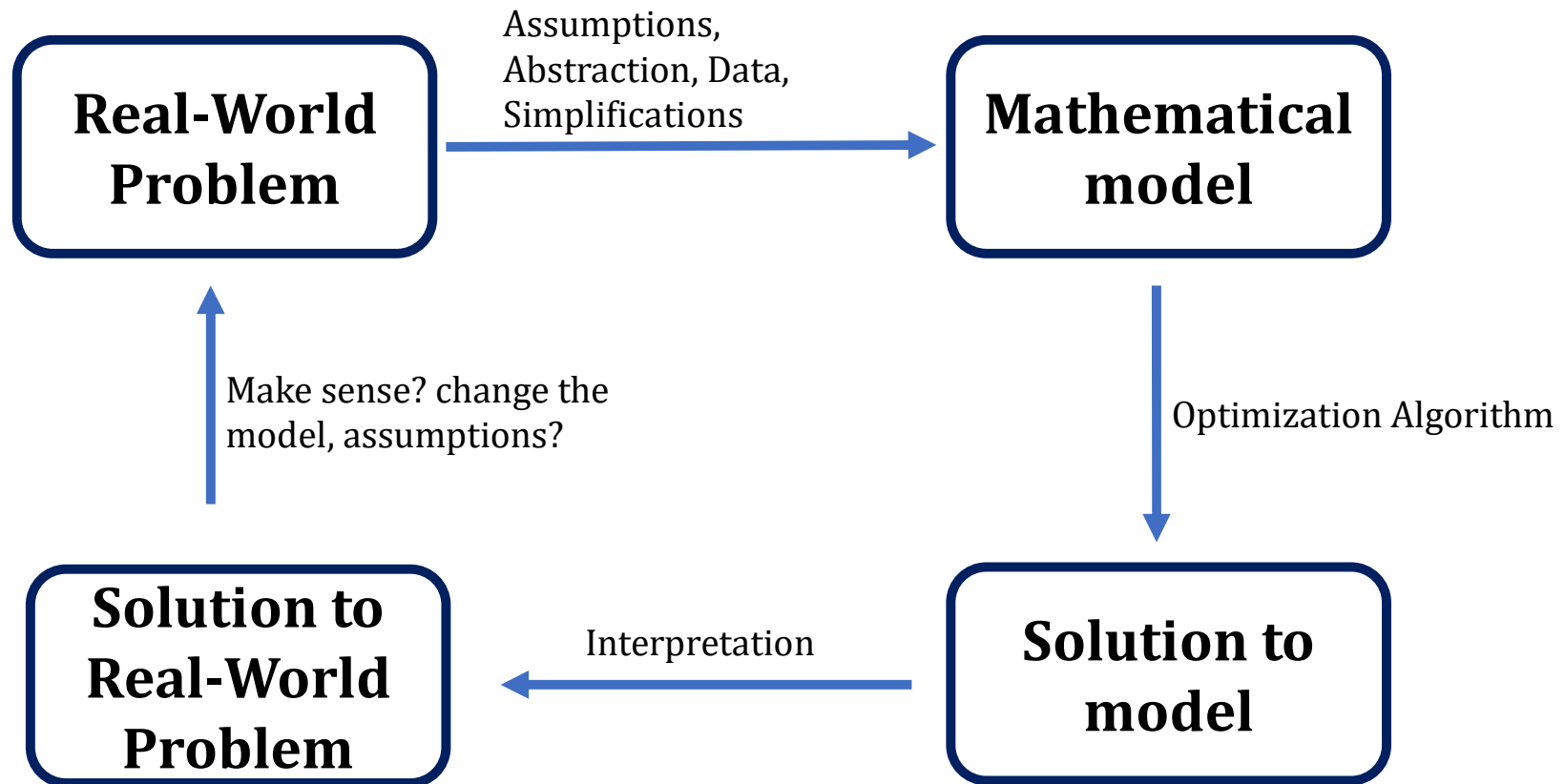
An Optimization problem seeks to find the largest (the smallest) value of a quantity (such as maximum revenue or minimum surface area) given certain limits to a problem.



An optimization problem can usually be expressed as find the maximum (or minimum) value of some quantity Q under a certain set of given condition.

Numerical Optimization

Schematic view of Optimization Process



Numerical Optimization

Mathematical Models in Optimization

Objective function

- An objective function is a formula involving one or more decision variables, denoted $x_1, x_2, \dots, x_n$ that is to be optimized by determining the appropriate values of those variables.
- It evaluates the **accuracy of the model's output**. The objective function is the measure used to evaluate how well the model's output matches the desired correct values.
- The objective function is a mathematical function that we want to **optimize or minimize/maximize in an optimization problem**. It represents the goal we are trying to achieve.
- In a typical **optimization problem**, we are looking to find the input variables that either maximize or minimize the objective function.
- The objective function can be linear or nonlinear, convex or non-convex, depending on the nature of the problem.
- Common types of objective functions include **cost functions, profit functions, utility functions**, etc.

Numerical Optimization

Mathematical Models in Optimization

The terminologies associated with Objective Function are as follows:

- **Constraints:** They are basically the conditional equations that govern the function.
- **Decision Variables:** The variable whose values are to be found out. The equations are solved so as to get the optimal value of these variables.
- **Feasible Region:** It is the region in the graph where the constraints are satisfied and the decision variables are found at the corners of the region.
- **Optimum Solution:** the best possible solution that satisfies all constraints and achieves the highest or lowest objective.
- **Infeasible Solution:** A solution that violates one or more constraints and cannot be implemented or executed.

Numerical Optimization

Mathematical Models in Optimization

Constraints and Constraint surface

In mathematical modelling, **a constraint is a condition or restriction that limits the values of variables or the behavior of a system.**

Constraints can be represented mathematically using equations or inequalities

Types of Constraints:

1. **Equality constraints:** Must be satisfied exactly (e.g., $x+y=10$)
2. **Inequality constraints:** Must be satisfied within a range (e.g., $x+y \leq 10$)
3. **Boundary constraints:** Limit the values of a variables (e.g., $0 \leq x \leq 10$)
4. **Non-negativity constraints:** Variables must be non-negative (e.g., $x \geq 0$)

Numerical Optimization

Mathematical Models in Optimization

Constraint surfaces:

In **two-dimensional** space, a constraint can be represented as a line or curve.

In **three-dimensional** space, a constraint can be represented as a surface. A constraint surface is a geometric representation of the constraint, showing the feasible region of solutions.

Example:

1. A constraint like $x+y=10$ can be represented as a line in 2D space.
2. A constraint like $x^2+y^2 \leq 0$ can be represented as a line in 2D space.

Constraint surfaces are useful in:

Optimization problems: to find the max or min of a function subject to constraints.

Decision-making: to Identify feasible solutions within the constraint boundaries

Modeling real-world systems: To account for physical or logical limitations.

Numerical Optimization

Mathematical Models in Optimization

The general form of an optimization model:

min or max $f(x_1, x_2, \dots, x_n)$ (Objective function)

Subject to **$g_i(x_1, x_2, \dots, x_n) \geq 0$** (Functional constraints)

$x_1, x_2, \dots, x_n \in S$ (set constraints)

$x_1, x_2, \dots, x_n$ are called decision variables

In words,

The goal is to find $x_1, x_2, \dots, x_n$ that

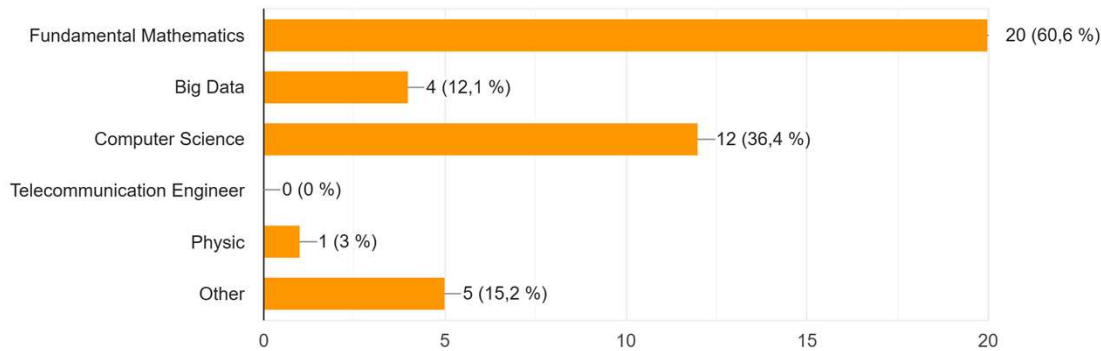
- Satisfy the constraints;
- Achieve min (max) objective function value.

Numerical Optimization

Presentation of Pre-course questionnaire results.

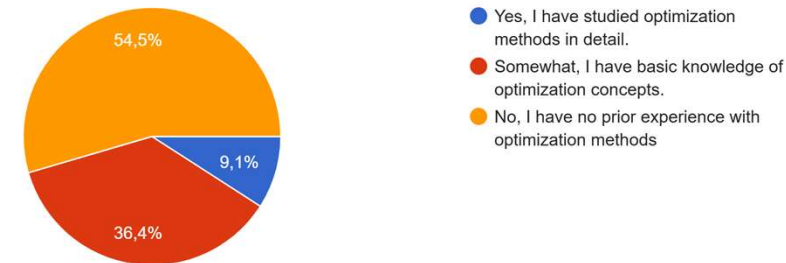
What is your primary area of study?

33 réponses



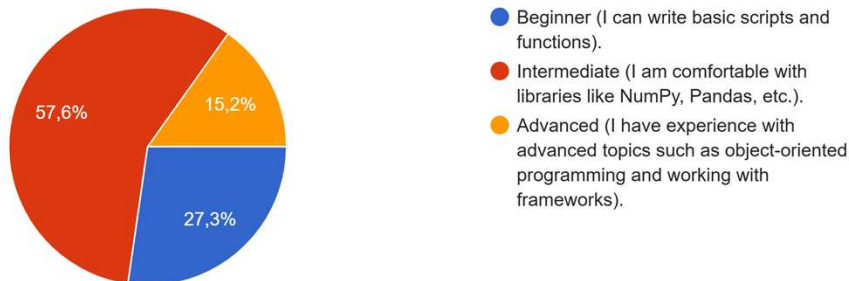
Do you have prior experience with numerical optimization ?

33 réponses



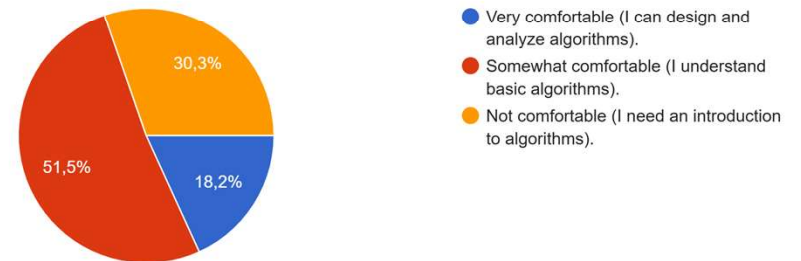
How would you rate your Python programming skills?

33 réponses



How comfortable are you with algorithms and computational problem-solving?

33 réponses

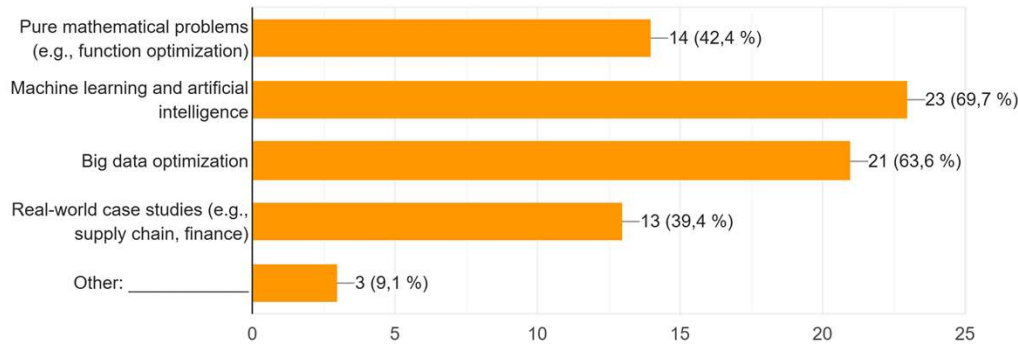


Numerical Optimization

Presentation of Pre-course questionnaire results.

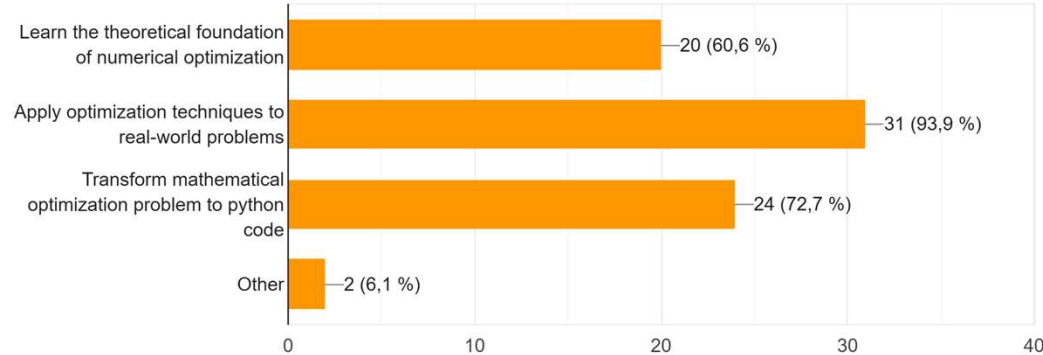
Which type of applications are you most interested in exploring during this course?

33 réponses



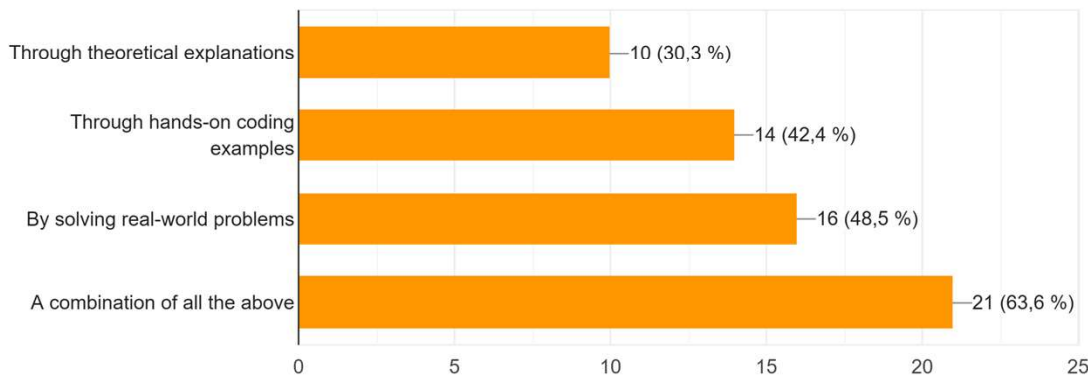
What are your primary goals for this course?

33 réponses



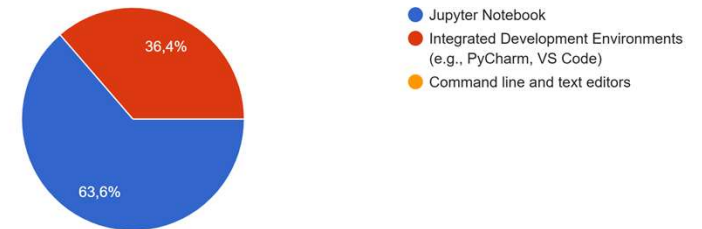
How do you prefer to learn new concepts?

33 réponses



What tools or environments do you prefer for Python programming?

33 réponses



Numerical Optimization

Presentation of Pre-course questionnaire results.

Problem Formulation as an Optimization Problem

Objective: Maximize the success rate of students in the Numerical Optimization Course, with an emphasis on achieving distinctions.

Problem Statement : Given a dataset of student responses from a pre-course questionnaire, we aim to:

- Identify clusters of students based on their responses.
- Optimize teaching strategies to address the specific needs of each cluster.
- Maximize the pass rate and the number of students achieving distinction.

Variables: Student characteristics (area of study, education level, prior optimization experience, programming skills, data visualization experience, linear algebra comfort, calculus comfort, algorithm comfort, learning preferences, etc.)

Numerical Optimization

Presentation of Pre-course questionnaire results.

Mathematical Formulation:

Let:

- x_i : Student i 's feature vector from questionnaire responses.
- C_k : Cluster k representing a group of students with similar characteristics.
- $P(C_k)$: Probability of passing for students in cluster C_k .
- $D(C_k)$: Probability of distinction for students in cluster C_k .

Objective:

$$\max \sum_{k=1}^K w_k \cdot P(C_k) + v_k \cdot D(C_k)$$

Subject to:

- $\sum_{k=1}^K w_k = 1$ (weights sum to 1).
- w_k and v_k are optimization weights for passing and distinction probabilities, respectively.

Numerical Optimization

Presentation of Pre-course questionnaire results.

Mathematical Formulation:

- **Unsupervised Learning for Clustering** : Group students with similar characteristics into clusters.
- **Objective Function**: Tailor teaching approaches per cluster to maximize $P(C_k)$ and $D(C_k)$
- **Constraints**: Time constraints, teaching resources, and curriculum structure.

The optimization approach will adjust weights w_k and v_k to ensure focus is given where it's most impactful.

Numerical Optimization

Presentation of Pre-course questionnaire results.

Programming Steps:

1.Input variables: Questionnaire responses.

2.Process:

1. Cluster students into similar groups.
2. Understand group strengths, weaknesses, and learning preferences.

3.Optimization: Design learning plans that increase success probabilities based on cluster needs.

4.Output: Improved pass rates and distinctions.

Example Cluster:

- Cluster 1: Students strong in theory but weak in coding.
- Cluster 2: Students comfortable with Python but lacking mathematical fundamentals.
- Cluster 3: Advanced students who can handle complex optimization problems.

OUTLINE

**Pre-Courses (
Python, Algebra,
Diff, Int etc.)**

**2. Interpolation and
Curve Fitting**

**3. Numerical
Optimization**



AIMS

**4. Algorithms for
Numerical
Optimizations**

**5. Real-World
Applications of
Numerical
Optimization**

6. Mini-Projects